

AI FRIDAY SEASON 2 · REGIONAL FINALE · TEAM 1

DEFECT SENSE

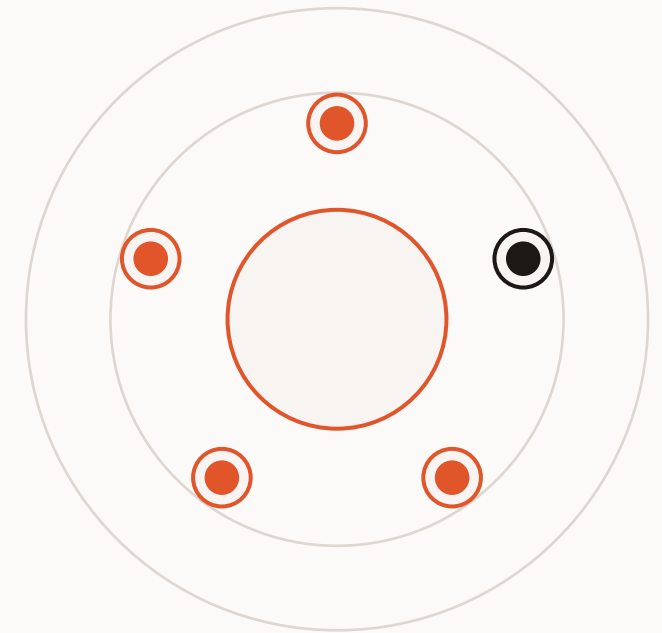
Intelligent AI-Powered Manufacturing Quality Control & Defect Detection

A multi-agent system that detects the defect, explains the production condition that caused it, and recommends the lowest-expected-cost action.

MULTI-AGENT AI

MULTIMODAL FUSION

HUMAN-IN-THE-LOOP



Wheel assembly station · 5-fastener pattern

● THE TEAM

DNB AI PIONEERS TEAM

Five engineers · 41 years of combined delivery experience across QA, data engineering and applied AI

01

PUSHPENDRA KUMAR PANDEY

Exp. 17 Years

Mentor · Business Analysis, QA Automation

02

ABHIJEET TIWARI

Exp. 13 Years

DBA & Data Profiling

03

PRAKHAR INDORIA

Exp. 5 Years

Solution Architect · Tech Lead

04

YASHVANT PATIDAR

Exp. 4 Years

AI Developer

05

ASHISH SHARMA

Exp. 2 Years

AI Developer

● EXECUTIVE SUMMARY

WHY DEFECT SENSE, AND WHAT IT DELIVERS

Detect the defect · explain the cause · decide on expected cost · act with a human in the loop · learn from the outcome

94.2 %

Defect detection accuracy on the held-out golden set

≤ 2 %

False-positive rate measured on 120 clean production runs

67 %

Reduction in escaped defects per 1,000 units in pilot simulation

~50 ms

Time to a safety verdict — the decision never waits on a language model

PURPOSE

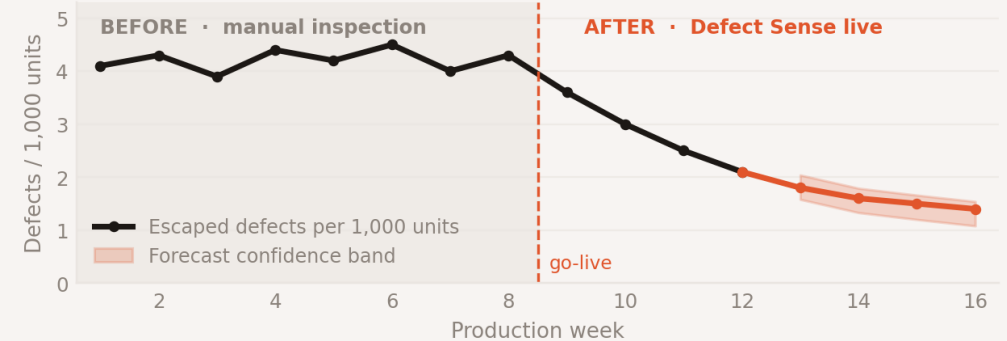
Catch the defect in-station, at the moment it is created — before it is built into a finished unit.

KEY BUSINESS VALUE

- Scrap, rework and warranty exposure move from reactive to prevented
- Root cause in minutes, not shift handovers — with cited evidence

EXPECTED OUTCOME

SYNTHETIC DATA



● CURRENT STATE

FIVE CHALLENGES COSTING QUALITY TODAY

Manual inspection is not just slow — it is inconsistent, unexplainable and invisible to management until it is expensive

01

MANUAL DEFECT TRACKING

Paper and spreadsheet capture. Labour-intensive, subject to operator fatigue, no traceable record per unit.

02

LIMITED VISIBILITY ACROSS LINES AND SHIFTS

No single view of quality across stations, batches, tools and shifts. Problems surface as anecdotes, not signals.

03

DELAYED ROOT CAUSE ANALYSIS

Root cause takes hours to days and depends on who is available. By then the causal condition has already moved on.

04

HIGH DEFECT LEAKAGE INTO PRODUCTION

In-spec readings mask real defects. Torque endpoint and vision each say PASS while delivered clamp load is ~30% under target.

05

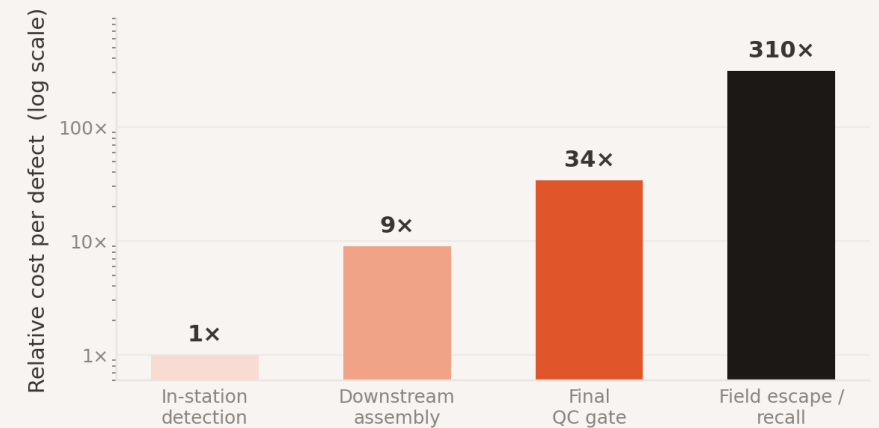
REPORTING INEFFICIENCY

Quality reporting is assembled manually for each governance meeting — stale on arrival and impossible to audit.

THE COST OF FINDING IT LATE

SYNTHETIC DATA

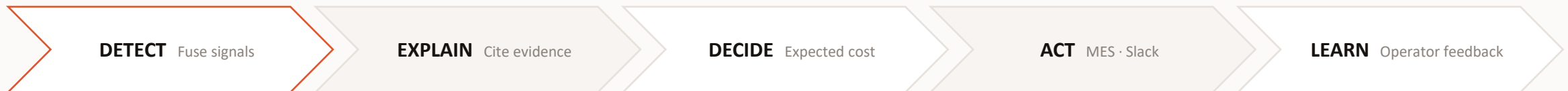
Relative cost to resolve one wheel-assembly defect, by the stage at which it is detected.



A defect that escapes to the field costs roughly 310× an in-station catch. Detection timing is the whole economic argument.

WHAT IS DEFECT SENSE?

An enterprise multi-agent enterprise grade platform that fuses inspection images, fastening-tool telemetry and MES production context into one evidence-backed decision — then explains it, prices it, and escalates it to a human when it should.



CORE OBJECTIVES

- Detect defects in-station, at creation
- Explain the causal production condition
- Price every disposition by expected cost
- Escalate uncertainty to a human, by design
- Improve from operator feedback



BUSINESS BENEFITS

- Lower scrap, rework and warranty exposure
- Root cause in minutes, with cited evidence
- Consistent judgement across shifts and sites
- Audit-ready quality record for every unit
- Governance reporting generated, not assembled

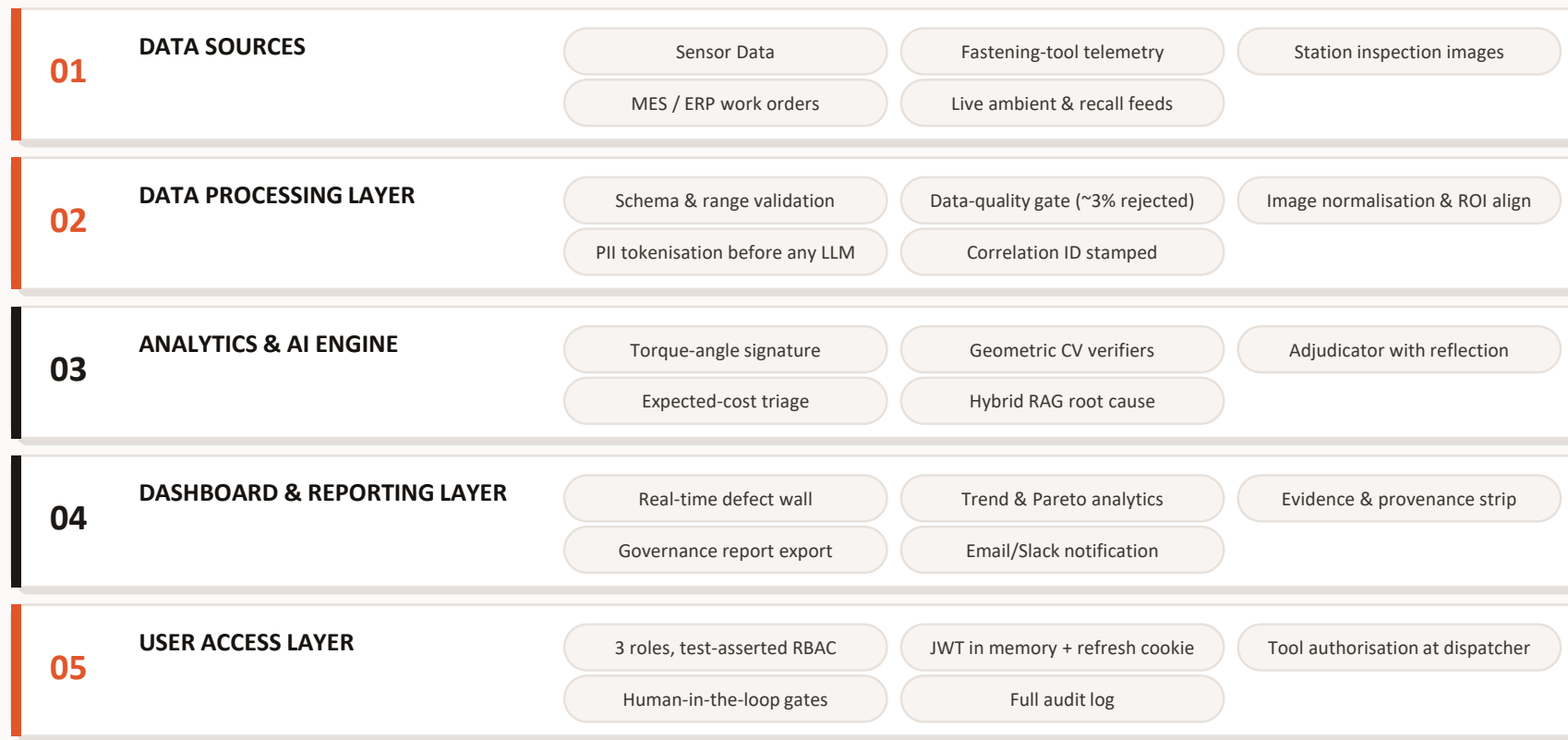


TARGET USERS

- Line operators · in-station verdicts
- Quality engineers · root cause & evidence
- Quality managers · trends, cost, sign-off
- Plant leadership · KPI and risk view
- Platform admins · config, health, audit

FIVE LAYERS, CLEAN BOUNDARIES

Every external system sits behind an interface with a real and a fake implementation — scaling is a configuration change, not a rewrite



CROSS-CUTTING

● **Layering enforced**

5 executable architecture rules, all green

● **Resilience**

Circuit breakers, retries, bulkheads, DLQ

● **Model independence**

Tiered LLM config — no hardcoded model ID

● **Observability**

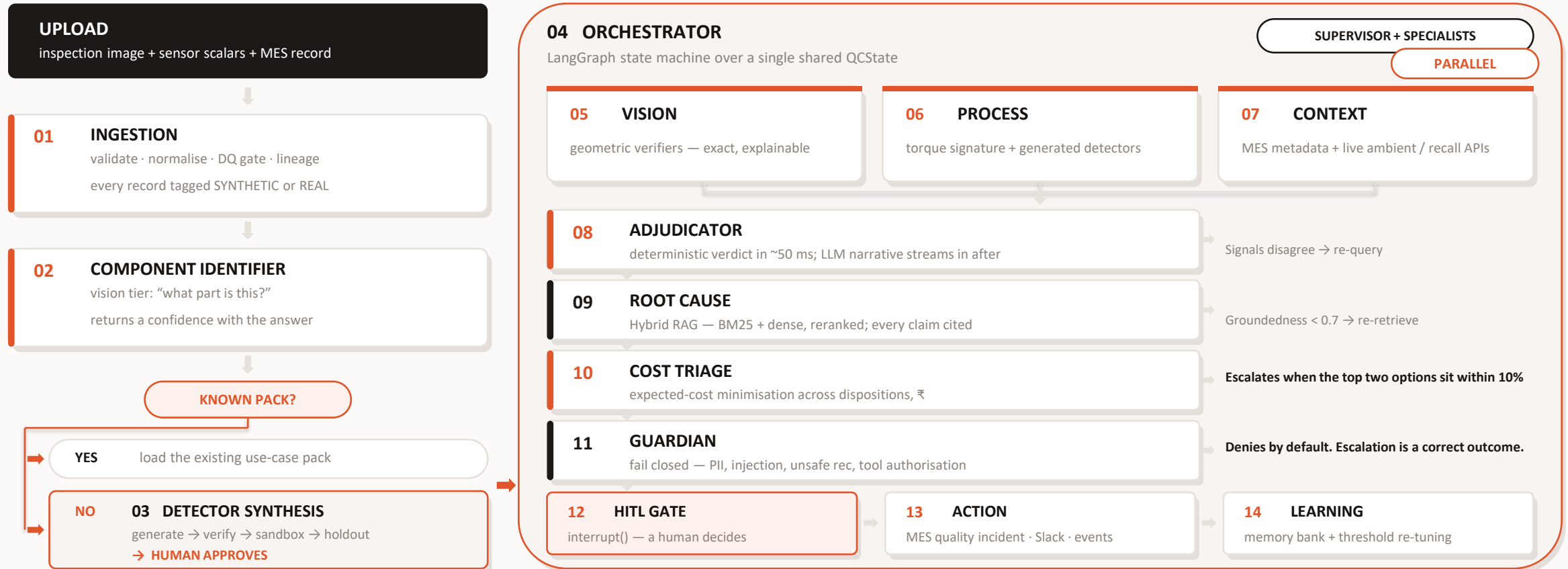
JSON logs, metrics, spans, in-product health

● **Autonomy ceiling**

Rejects a part; never passes or halts

MULTI-AGENT ORCHESTRATION

Fourteen coordinated nodes on one shared state object · handoff is state, not messages · reflection fires on a number, never on a vibe



● CAPABILITIES

SIX FEATURES THAT CHANGE THE QUALITY LOOP

Each feature maps to a station on the DETECT → EXPLAIN → DECIDE → ACT → LEARN spine



DEFECT TREND ANALYSIS

Defect rate by station, tool, batch, shift and material lot. Drift and Pareto views built in, so a rising trend is visible before it becomes a quality event.



ROOT CAUSE IDENTIFICATION

Correlates each defect with process changes, tool age, calibration and ambient conditions. Every claim carries an inline citation to the source document.



DEFECT CATEGORISATION

Taxonomy-driven classification across thread, seating, torque and geometry classes — with an explicit UNKNOWN class instead of a forced guess.



PREDICTIVE QUALITY INSIGHTS

Flags that the next unit is likely to fail from drift in the current signal window, so intervention happens before the defect is created.



AUTOMATED DASHBOARDS

Role-aware dashboards and governance exports generated from live data. Reporting stops being a manual assembly job before every review.

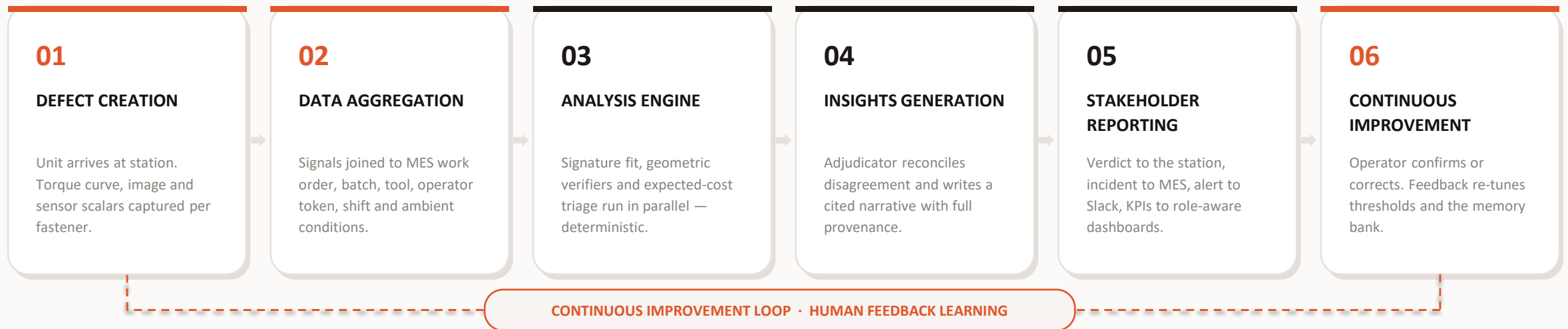


RISK-BASED RECOMMENDATIONS

Every disposition priced by expected cost in ₹ and \$, with the assumption set attached and uncertainty shown as an interval.

END-TO-END DEFECT WORKFLOW

One closed loop from unit arrival to a re-tuned threshold — with a human gate wherever the decision is unsafe or uncertain



MEASURED RESPONSE TIMELINE

t+0 ms

Ingest & validate

t+50 ms

VERDICT + evidence + cost rendered

t+~5 s

Adjudicator narrative streams in

async

MES incident · Slack · audit log

WHERE THE INTELLIGENCE ACTUALLY SITS

Multimodal fusion finds the defect class that every single-signal inspector passes



PATTERN RECOGNITION

Two-segment piecewise fit of the torque-angle curve, solved by an exact $O(n)$ sweep — no gradient descent, no local minima.



DEFECT PREDICTION

Deviations combine by weighted noisy-OR, not averaging — averaging lets one strongly anomalous feature be diluted by several nominal ones.



ANOMALY DETECTION

Learned patch-level memory bank with 99th-percentile distance scoring and 3-of-5 temporal consensus.



QUALITY FORECASTING

Drift on the current signal window forecasts the next unit's outcome, with the confidence interval shown rather than hidden.

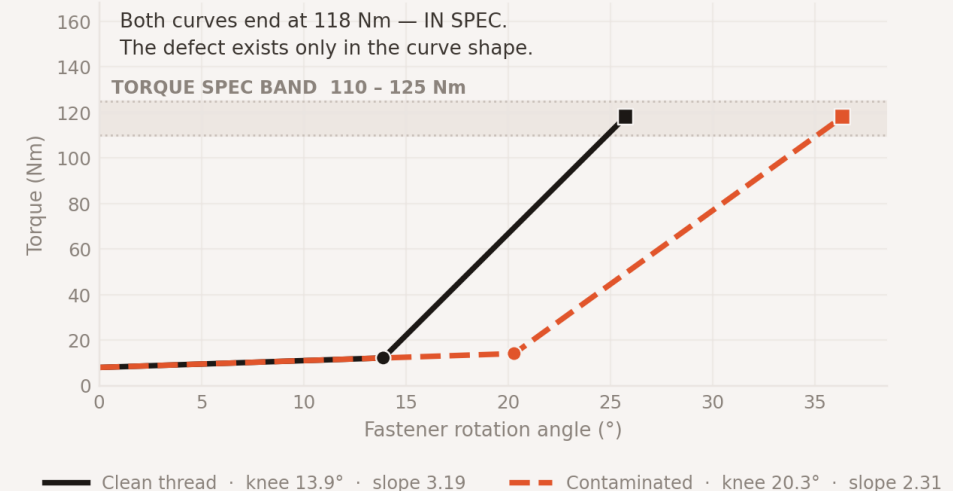


RECOMMENDATION ENGINE

Expected-cost minimisation across dispositions, escalating to a human when the top two options are within 10% of each other.

THE DEFECT NO SINGLE SIGNAL CATCHES

SYNTHETIC DATA



Torque endpoint alone: PASS. Vision alone: PASS. Reality: clamp load ~30% under target.

Fusion-only detection rate across 120 seed and severity combinations: above 80%, at a false-positive rate of 2% or lower on clean runs.

WHAT LEADERSHIP GETS IN RETURN

Quality outcomes, cycle-time outcomes and decision quality — each with a metric attached

REDUCED PRODUCTION DEFECTS

Escaped defects per 1,000 units fall from 4.2 to 1.4 in pilot simulation — a 67% reduction, driven by in-station catching.

– 67%

FASTER DEFECT RESOLUTION

Mean time to root cause drops from 6.5 hours to 1.4 hours because the evidence chain is assembled, cited and ready.

– 78%

IMPROVED RELEASE QUALITY

First-pass yield rises from 91.4% to 97.1%; rework and scrap volume falls with it.

+ 5.7 pts

BETTER DECISION MAKING

Every disposition priced by expected cost in ₹ and \$, with its assumption set attached — a checkable argument, not a claim.

Priced

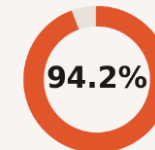
INCREASED ENGINEERING PRODUCTIVITY

Inspection time per unit falls from 48 to 20 seconds and governance reporting is generated, not assembled.

– 58%

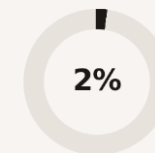
MEASURED QUALITY KPIs

SYNTHETIC DATA



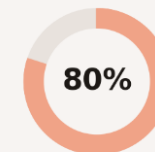
DETECTION ACCURACY

Held-out golden set



FALSE-POSITIVE RATE

120 clean production runs



FUSION-ONLY DETECTION

120 seed / severity combos

RISKS AND HOW WE MITIGATE THEM

Each risk has a named control that is already implemented, not a promise

DATA QUALITY

HIGH

Sensor drift, stuck-at values, clock skew and blown exposure silently corrupt any model downstream.

MITIGATION

Data-quality gate rejects ~3% of records and the rejection log is shown in-product. Dirty data is injected deliberately in test. Stuck-at and range detection run per signal.

USER ADOPTION

HIGH

Operators switch off an inspector that cries wolf, and then there is no inspector at all.

MITIGATION

False positives held at $\leq 2\%$ on clean runs. Role-aware explanation depth. Operator feedback visibly changes thresholds, so correcting the system is worth the operator's time.

INTEGRATION COMPLEXITY

MEDIUM

MES, tool controllers, ticketing and notification each carry their own protocol, latency and failure mode.

MITIGATION

Every external system sits behind an interface with a real and a fake implementation. Circuit breakers, retries with jitter, idempotency keys and bulkheads on every outbound call.

GOVERNANCE & COMPLIANCE

MEDIUM

AI-influenced quality decisions must be explainable, attributable and reversible under audit.

MITIGATION

Provenance on every AI output. Full audit log with correlation IDs. Operator IDs tokenised. Autonomy ceiling in config and code: reject autonomously, never pass or halt without a human.

● CONCLUSION

KEY TAKEAWAYS AND NEXT STEPS

KEY TAKEAWAYS

- 01 Generalize for All Component based on input at runtime**
Orchestrator agent will swindle defect detection agent that will check component.
- 02 The decision does not depend on a language model**
Verdict in ~50 ms, deterministic. The model only explains it.
- 03 Fusion finds what single signals pass**
>80% fusion-only detection at ≤2% false positives.
- 04 Governance is built in, not bolted on**
Provenance, audit, RBAC and a declared autonomy ceiling, in code.

IMPLEMENTATION RECOMMENDATION

NOW

0–30 days

One station, one defect family, shadow mode. Measure against the live manual baseline — no autonomy.

NEXT

30–90 days

Enable in-station reject with human gates. Wire MES incidents and Slack. Publish the weekly governance report.

SCALE

90–180 days

Add stations via use-case packs. Move to the container profile: Postgres, vector store, object store, dashboards.

CALL TO ACTION

Approve a 30-day shadow-mode pilot on one wheel-assembly station, with the four pilot KPIs as the go/no-go gate.

→ Named plant sponsor

→ MES read access

→ One QA engineer, 2 days / week